

Notes on lecture 6

Jiten Chandra Kalita and Shyamashree Upadhyay

1 Tangent planes continued ...

Recall that the equation of the tangent plane to a surface $z = f(x, y)$ at the point (a, b, c) (where $c = f(a, b)$) is given by

$$z - c = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Remark 1.0.1. In order that the tangent plane exists at the point (a, b, c) , the first order partial derivatives f_x and f_y must be continuous at (a, b) . Otherwise, we won't be able to draw the tangent plane!

Question: If f is not continuous at (a, b) , can we draw a tangent plane to the surface $z = f(x, y)$ at (a, b, c) , where $c = f(a, b)$?

Answer: No. In fact, even if f is continuous at (a, b) , there may be some problems with being able to draw a tangent plane. For being able to draw a tangent plane at the point (a, b, c) to the surface $z = f(x, y)$, we must have that the first order partial derivatives f_x and f_y must be continuous at (a, b) .

Example 1.0.2. Consider the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Figure 1.0.1 shows the graph of f .

Observe that $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist (because if we take the limit along the line $y = mx$, the value of the limit will be $\frac{m}{1+m^2}$, which varies as m varies). Since the limit does not exist, therefore f is not continuous at $(0, 0)$. Again,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

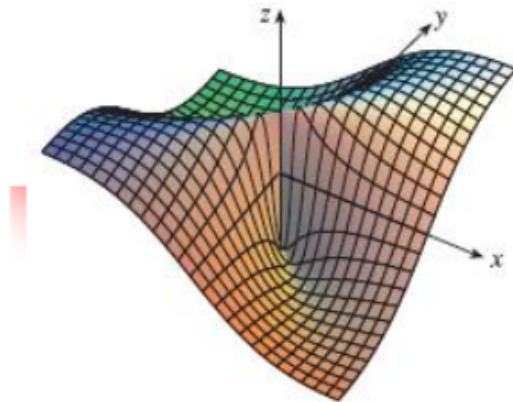


Figure 1.0.1: Figure for Example 1.0.2

Similarly, $f_y(0, 0) = 0$.

Though f is not continuous at $(0, 0)$, but the partial derivatives f_x and f_y exist at $(0, 0)$. Now the question is that can we draw a tangent plane to the surface at the origin? For this, let us check whether f_x and f_y are continuous at $(0, 0)$ or not.

Observe that

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f_x(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{y^3 - x^2y}{(x^2 + y^2)^2} \\ &= \lim_{(x,y) \rightarrow (0,0) \text{ along } y\text{-axis}} \frac{y^3}{y^4} = \lim_{y \rightarrow 0} \frac{1}{y}, \end{aligned}$$

and this limit does not exist. Similarly, $\lim_{(x,y) \rightarrow (0,0)} f_y(x, y)$ does not exist. So f_x and f_y are not continuous at the origin. That's why we won't be able to draw a tangent plane to the surface under consideration at the origin. $\square$

Definition 1.0.3. Let f be a function of two variables x and y . Let (a, b) be a point in the domain of f . If the partial derivatives f_x and f_y exist at (a, b) and are also continuous at (a, b) , then we say that f is **continuously differentiable** at (a, b) . $\square$

2 Inward and outward normal

For any given surface and a point on it, we can define the normal at that point (to the surface) to be either inward or outward, depending upon the orientation of the surface.

Consider an arbitrary vector in space. If we want to find out the orientation of the vector, then we will have to check whether the vector is going to lie above the xy -plane or below the xy -plane. If the vector is going to lie above the xy -plane, then its **orientation** will be **upwards**. And if the vector is going to lie below the xy -plane, then its **orientation** will be **downwards**. If the components of the vector are p, q, r , that is, if the vector is $\langle p, q, r \rangle$, then we will get an upward orientation if $r > 0$, and downwards if $r < 0$.

In the last lecture, we had got that the normal $\vec{n}$ to the surface $z = f(x, y)$ at the point (a, b, c) (where $c = f(a, b)$) is given by

$$\vec{n} = \langle f_x(a, b), f_y(a, b), -1 \rangle.$$

Since the 3rd component of $\vec{n}$ is -1 here, which is < 0 , so the orientation of $\vec{n}$ is downwards. So to get a normal vector having upward orientation, we must take $-\vec{n}$. See Figure 2.0.1 for an illustration.

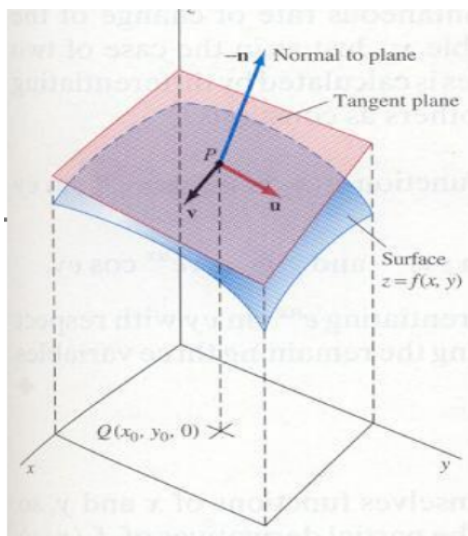


Figure 2.0.1: The upward normal is $-\vec{n}$

In the last lecture, we had seen that the equation of the tangent plane to the

surface $z = f(x, y)$ at the point (a, b, c) (where $c = f(a, b)$) is given by

$$f_x(a, b)(x - a) + f_y(a, b)(y - b) - (z - c) = 0.$$

The above equation can also be written as

$$-f_x(a, b)(x - a) - f_y(a, b)(y - b) + (z - c) = 0.$$

From the last equation, we can say that the normal vector is given by $\langle -f_x(a, b), -f_y(a, b), 1 \rangle$, which is nothing but $-\vec{n}$.

But it is not always true that upward orientation means outward orientation and downwards means inwards. Consider the sphere $x^2 + y^2 + z^2 = a^2$. If we consider the lower hemisphere, then we can easily see that the outward direction of the normal at any point on it is going to be downwards (as the 3rd component of the outward normal vector is going to be negative). Similarly, if we consider the upper hemisphere, then the outward direction is going to be upwards. So in order to decide whether a given normal vector is **outward** or **inward**, we will have to look at the surface itself. For instance, for the lower hemisphere, the inward normal is going to have an upward direction. Basically, the concept of inward/outward normal is going to depend upon the surface.

3 Linear approximation

Let us begin this section with a motivating example.

Example 3.0.1. Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

In Figure 3.0.1, we can see that as we move closer and closer to the point $(1, 1, 3)$, the surface is looking flatter and flatter. Gradually, in part (c) of Figure 3.0.1, we can see that the tangent plane is almost overlapping with the surface.

In Figure 3.0.2, we can see the contour lines of the surface $z = 2x^2 + y^2$. They are obtained by slicing the surface with some plane parallel to the xy -plane. Observe that the curvature of the contour lines is decreasing as we are going nearer to the point $(1, 1)$.

Observe that here we have $f(x, y) = 2x^2 + y^2$. We are considering a tangent

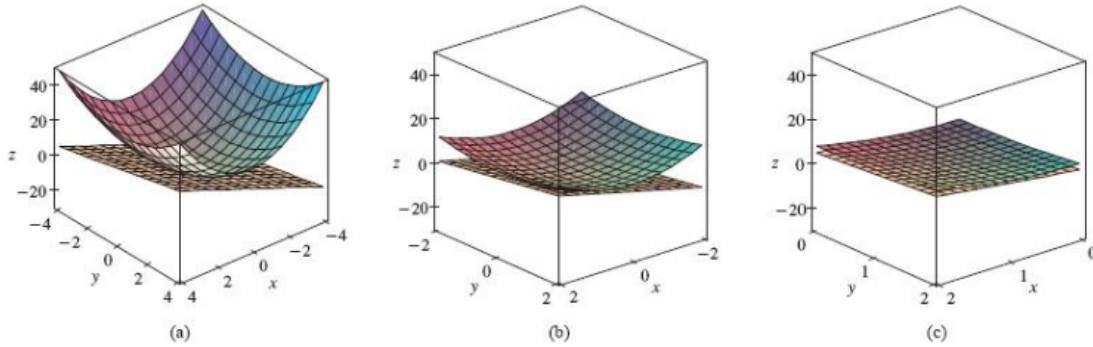


Figure 3.0.1: Tangent plane to $z = 2x^2 + y^2$ at $(1, 1, 3)$

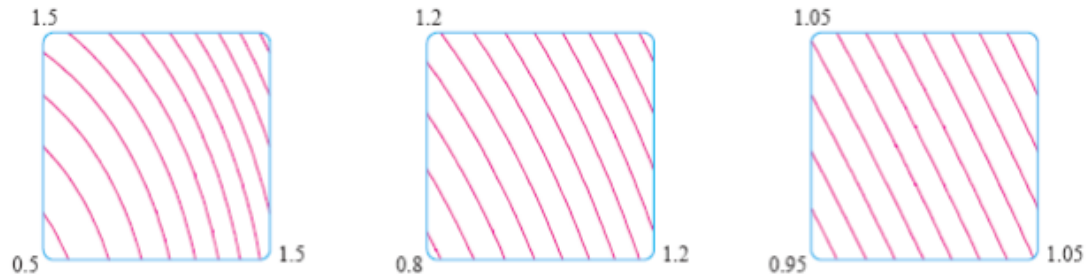


Figure 3.0.2: Contour lines of the surface $z = 2x^2 + y^2$

plane to the surface $z = f(x, y)$ at the point $(1, 1, 3)$. So $f_x = 4x$ and $f_y = 2y$. This implies that $f_x(1, 1) = 4$ and $f_y(1, 1) = 2$. The equation of the tangent plane is then given by

$$z - 3 = 4(x - 1) + 2(y - 1).$$

Or in other words, the equation of the tangent plane is

$$z = 4x + 2y - 3.$$

Let $L(x, y) = 4x + 2y - 3$. We have extracted the function $L(x, y)$ using the equation of the tangent plane to the surface corresponding to $z = f(x, y)$. Let us take a point on the xy -plane, which is very near to the point $(1, 1)$,

say $(1.1, 0.95)$. Observe that

$$L(1.1, 0.95) = 3.3 \text{ and } f(1.1, 0.95) = 3.3225.$$

So the values of L and f are very close to each other at $(1.1, 0.95)$. But if we consider the point $(2, 3)$, which is a little away from the point $(1, 1)$, then we see that

$$L(2, 3) = 11 \text{ and } f(2, 3) = 17.$$

The values of L and f are not that much close to each other at $(2, 3)$.

This means that if we want to approximate the function f with the help of the new function L (obtained via the tangent plane procedure) in the vicinity of the point at which the tangent plane has been considered, then we get a very good approximation. But if we move away from the point at which the tangent plane has been considered, the values are going to deviate. In that case, we no longer have a good approximation. $\square$

The approach given in Example 3.0.1 above is called the **linear approximation** or the **tangent plane approximation**. It is clear that if we consider the tangent plane to a surface given by some function at a particular point, then we will always get a linear function (because x and y are always going to appear linearly in the equation of the tangent plane). That is why this approach is called the **linear approximation**.

With reference to Example 6.0.1 above, the function L is called the **linearization** of f at $(1, 1)$, and the approximation

$$f(x, y) \approx 4x + 2y - 3$$

is called the **linear approximation** or the **tangent plane approximation** of f at $(1, 1)$.

In general terminology, we know that the tangent plane to the graph of a function f of 2 variables at the point $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The linear function whose graph is this tangent plane, namely,

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linearization** of f at (a, b) and the approximation

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linear approximation** or the **tangent plane approximation** of f at (a, b) .

Question: What is the need for a linear approximation?

Answer: We are getting a very simple looking function $L(x, y)$ out of the procedure of linear approximation. Sometimes, the function f under consideration may be very complicated. Then by this approach of linear approximation, we can simply approximate the complicated function f .

4 Differentiability

4.1 For a function of one variable

Consider the function $y = f(x)$ at $x = a$, where a is a point in the domain of f . Let Δx denote a small increment along the x direction. Let

$$\Delta y = f(a + \Delta x) - f(a).$$

Let

$$f'(a) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

We say that f is **differentiable at a** if Δy can be expressed in the form

$$\Delta y = f'(a)\Delta x + \epsilon\Delta x,$$

where $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$.

Recall from single variable calculus that $f'(a)$ is the slope of the tangent to the curve $y = f(x)$ at the point $(a, f(a))$.

4.2 For a function of 2 variables

Let f be a function of 2 variables x and y . Let (a, b) be a point in the domain of f . Let $z = f(x, y)$. Let Δx and Δy be small increments along the x and y directions respectively. Define

$$\Delta z = f(a + \Delta x, b + \Delta y) - f(a, b).$$

We know that the equation of the tangent plane to the surface $z = f(x, y)$ at the point $(a, b, f(a, b))$ is given by

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$= f_x(a, b)\Delta x + f_y(a, b)\Delta y.$$

Whereas, for a function of one variable, the equation for the tangent line was given by

$$y - f(a) = f'(a)(x - a).$$

We say that $f(x, y)$ is **differentiable at** (a, b) if Δz can be expressed in the form

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y,$$

where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

Remark 4.2.1. In case of a function of one variable, we had considered the tangent line approximation. Here, for the case of a function of 2 variables, we consider the tangent plane approximation. The geometrical interpretation of this can be seen in Figures 4.2.1 and 4.2.2. Note that in Figure 4.2.1, $\Delta y = y - f(a)$, and in Figure 4.2.2, $\Delta z = z - f(a, b)$.

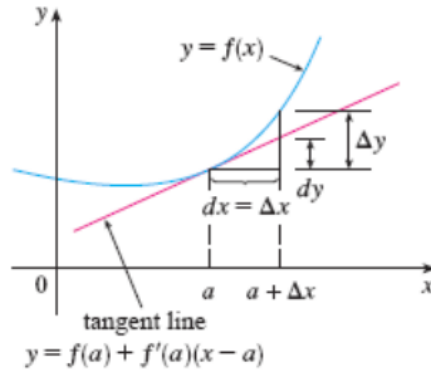


Figure 4.2.1: The tangent line approximation

Theorem 4.2.2. *If f is differentiable at (a, b) , then it is continuous at (a, b) .*

PROOF: From the definition of differentiability, we have

$$f(a + \Delta x, b + \Delta y) - f(a, b) = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y,$$

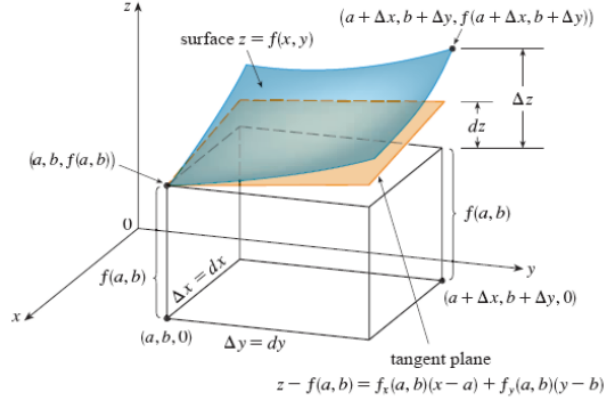


Figure 4.2.2: The tangent plane approximation

where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

Observe that as $(\Delta x, \Delta y) \rightarrow (0, 0)$, the right hand side of the above equation is going to tend to zero. So we have

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} f(a + \Delta x, b + \Delta y) = f(a, b).$$

This shows that f is continuous at (a, b) . □

Example 4.2.3. Show that the function defined by $f(x, y) = (x^{1/3} + y^{1/3})^3$ is continuous and has partial derivatives at the origin, but not differentiable. It can be easily seen that f is continuous at $(0, 0)$ (think why?). Let $\Delta x = h$ and $\Delta y = k$. Then

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1$$

and similarly, $f_y(0, 0) = 1$.

From the definition of differentiability of f at the point $(0, 0)$, we have

$$f(h, k) - f(0, 0) = f_x(0, 0)h + f_y(0, 0)k + \epsilon_1 h + \epsilon_2 k.$$

That is,

$$(h^{1/3} + k^{1/3})^3 = h + k + \epsilon_1 h + \epsilon_2 k.$$

For f to be differentiable at $(0, 0)$, the above equality should hold true, where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $(h, k) \rightarrow (0, 0)$.

The above equality can be rewritten as

$$h + k + 3h^{2/3}k^{1/3} + 3h^{1/3}k^{2/3} = h + k + \epsilon_1 h + \epsilon_2 k.$$

In other words, it is the same as

$$3 \left(\frac{k}{h} \right)^{1/3} h + 3 \left(\frac{h}{k} \right)^{1/3} k = \epsilon_1 h + \epsilon_2 k.$$

So we must have:

$$\epsilon_1 = 3 \left(\frac{k}{h} \right)^{1/3}$$

and

$$\epsilon_2 = 3 \left(\frac{h}{k} \right)^{1/3}.$$

Now we know that for f to be differentiable at $(0, 0)$, we must have $\epsilon_1, \epsilon_2 \rightarrow 0$ as $(h, k) \rightarrow (0, 0)$. But here, if $(h, k) \rightarrow (0, 0)$, then ϵ_1 and ϵ_2 are not well defined!

So f is NOT differentiable at $(0, 0)$, although both f_x and f_y existed at $(0, 0)$. It is an **exercise** for you to check that f_x and f_y are not continuous at $(0, 0)$.

□

The example given in Figure 4.2.3 is a **homework** for you:

Example Show that the function defined by

$$f(x, y) = \begin{cases} x^2 \sin(\frac{1}{x}) + y^2 \sin(\frac{1}{y}) & \text{if } xy \neq 0 \\ x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0, y = 0 \\ y^2 \sin(\frac{1}{y}) & \text{if } x = 0, y \neq 0 \\ 0 & \text{if } x = y = 0 \end{cases}$$

is differentiable at the origin, but not continuously differentiable thereat because $f_x(x, y)$ is not continuous there.

Figure 4.2.3: Example for homework

Theorem 4.2.4. *If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .*

PROOF: Omitted. □

Remark 4.2.5. The converse of Theorem 4.2.4 is NOT necessarily true. To see this, please work out the example given in Figure 4.2.3.

Please take the examples given in Figure 4.2.4 also as your **homework**.



Example: Show that the function f , where

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & \text{if } x^2 + y^2 \neq 0 \\ 0 & \text{if } x = y = 0 \end{cases}$$

is differentiable at the origin.

Example: Consider the function f , where

$$f(x, y) = \begin{cases} x^2 \sin \frac{1}{x} + y^2 \sin \frac{1}{y} & \text{if } xy \neq 0 \\ x^2 \sin \frac{1}{x} & \text{if } x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y} & \text{if } x = 0, y \neq 0 \\ 0, & \text{if } x = y = 0 \end{cases}$$

Is the function continuously differentiable at the origin? What about its differentiability thereat?

Figure 4.2.4: Examples for homework

4.3 For a function of n variables

Let V_n denote the n -dimensional space. A function $f(\mathbf{x})$ is said to be **differentiable** at a point $\mathbf{a} \in V_n$ if there exists a vector $\mathbf{C} \in V_n$ such that

$$\lim_{\mathbf{h} \rightarrow 0} \frac{f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - \mathbf{C} \cdot \mathbf{h}}{|\mathbf{h}|} = 0,$$

where $\mathbf{h} \in V_n$.

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